Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India
(Autonomous College of Affiliated to University of Mumbai)

End Semester Examination
Dec.2024

Max. Marks: 100 **Duration: 3 Hrs**

Class: TE **Semester: V**

Course Code:CS301/AI301/DS301 **Program: CE/CSE**

Name of the Course: Theory of Computation

Instruction:
(1) All questions are compulsory
(2) Draw neat diagrams and write equations wherever necessary
(3) Assume suitable data if necessary

Course Outcomes at the end of the course student will be able to:
(1) Design a finite automaton for regular expressions and languages.
(2) Apply the properties of regular languages.
(3) Construct the grammar for a language and convert it into normal forms.
(4) Design and evaluate pushdown automata and Turing Machine for a language.

Q No.		Max. Marks	CO BL PI
Q.1a	Construct a CFG for the language consisting of all binary strings in which 011 does not appear as a substring.	10	3-3
	OR		
	Construct a CFG for the language consisting of all binary strings in which the number of 0's is equal to number of 1s	10	3-3
Q.1b	Give an example of a deterministic Turing Machine that does not halt (loops forever) on at least one input.	10	4-2
Q.2a	Give the full formal description (using the state diagram) of a deterministic TM that accepts all the strings in language $\{a^n \mid n \geq 1$ and n is a power of 2$\}$ and rejects everything else.	10	4-3

Q2.b	Give a non-deterministic push down automaton for the language consisting of all strings over the alphabet {a,b}. Which have odd length and whose middle alphabet is a.	10	4-3
Q.3a	Give a CFG for the language consisting of all binary strings that begin as well as end with a 0.	10	3-2
Q.3b	Design a finite state machine to add two binary numbers of same length.	10	1-3
Q.4a	Design a finite automaton to accept set of all strings that does not contain "00" over the input (0,1) and construct the regular expression for the same using Arden's theorem.	10	1-3
Q.4b	Prove that the language $L = \{a^i b^{2i} \mid i > 0\}$ is not regular by applying pumping lemma. Clearly explain the conditions of lemma and provide step-by-step justification of your proof.	10	2-3
Q.5a	Simplify the following grammar: $S \rightarrow aS \mid bA$ $A \rightarrow bA \mid B \mid \varepsilon$ $B \rightarrow a \mid \varepsilon$ **OR** Reduce the following production of grammar to Greibach Normal Form $S \rightarrow ABC$ $A \rightarrow a \mid b$ $B \rightarrow Bb \mid aa$ $C \rightarrow aC \mid CC \mid ba$	10 10	3-3 3-3
Q.5b	Show that the language consisting of all binary strings divisible by 6 is regular. (Hint: if two languages L1 and L2 are regular then so is their union)	10	2-3